

1.Images and evidence should be included throughout.

Manager should make a copy, share with each member of the group and (jmiranda@rpsk12.org, pnguyen@rpsk12.org), make it viewable by “ANYONE with the link” and submit it to Google Classroom when complete. Delete the notes in red when you are done with each section.

Group Members:

Element A: Presentation and Justification of the Problem

Item	Notes/Links/Details
Problem Statement: Not enough freshman students are able to learn/memorize the quadratic formula. We think this is a problem because not enough students are engaged in the lesson which then causes them to be less encouraged to learn and memorize the quadratic formula. We want to make something that freshman students actually want to learn from while having fun and then they would be more encouraged to memorize it. Identify the need. Move past describing the situation to describe how it is a problem. Make sure that the problem is explained in enough detail that anyone could understand the goal of this engineering design problem. Assume your reader knows nothing about the problem or its possible solutions. Include details that you might consider common knowledge. (How is it a problem and why it is a problem)	To make the quadratic formula easier to memorize or learn and make students more engaged
Problem Definition: The quadratic formula is $ax^2 + bx + c = 0$ Use your research to describe your problem in detail. In this element, focus on depth, not breadth. Depth means digging down into the problem, pulling out details that aren't noticeable at first. Define the problem by incorporating relevant research to get a “3” rating. Who is involved? What are they doing? Where is this happening? When does it occur? Why does it happen? How often does it happen? What unique situations does it occur within, if any? What is the size of the issue? The people that are affect by this is the students to a level because they are not learning the quadratic formula.The teacher could also benefit from this because it will make their job easier (what type of math, what type of learning)	

Additional information about the problem: How does this problem relate to other contexts? Can solving this problem impact other stakeholders? How can this problem have consequences in other situations? Students need more effective ways of learning. Students are not engaged in the things that they are being taught. This will create bad habits for students because they won't know if they are interested in something if they aren't engaged at the start. This also affects people looking for jobs. They might not know what they like or have that much experience	
Stakeholders: Identify primary and secondary stakeholders and their specific roles in the problem. In order to earn a score of 3, you will need to identify several (more than three) primary stakeholder groups. Teachers (Josh and Peter) Students (Vitoria (freshman) and Sergio (senior))	
Concerns of Stakeholders: Highlight the concerns of several (more than three) primary stakeholder groups. You are encouraged to use cause and effect phrasing: "Due to <insert stakeholder concern> there is a need to _____. To start this, identify questions you want to ask math teachers. Create a google form to ask those questions. Peter and I will assist in getting responses. Come up with questions and a way to ask them for your other stakeholders. You will update this section when you receive responses. Josh: Students need more hands on learning. Josh thinks students should learn the skill, not the ideas! He also thinks students should find something they are interested in so they can learn more effectively. Peter: says students think that the lessons are not important. Students also think that teachers should just provide the answers instead of teaching them. Students could understand the usefulness in quadratics and find quadratics in day to day things and games such as basketball. Teachers and student energy rub off on each other: if the teacher is tired students could be more tired. Vitoria freshman: Says "It's not that I can't learn it I just can't remember" suggests students should have a review of everything they have learned "Not a test but review day." Lots of distractions "Cell Phones and Chromebook" also says that students just start to give up when they don't understand something. Students should have about a 5 minute break so they could get all of their chatter out and phone usage. Sergio senior: Students have lots of distractions ex: phones. Students are unaware about freshman year and its impact on students' high school years. Make the lesson more fun. Try to get more group work because group work is "Funnier". Put students who actually care about math in a group and put the students who don't care as much as math in a group. Some students have a hard time visualizing the problems.	

Problem Justification: Use in-line citations to explicitly state which information is from which source. Do not just list all of the citations at the end of the submission or in footnotes. Cite at least 3 current, credible sources to get a "3" rating. These are the sources that you used to find data/statistics, the research to justify your problem statement, and the interview with the stakeholder(s). Use data and/or statistics to back up your argument (i.e. sources): "Previous studies by _____ show that _____." Explain why it is important to solve this problem now with measurable data. Include data and statistics that helps the reader understand the severity of the problem. Convince your readers that the problem needs to be solved.

"Previous studies by interviewing students and teachers show that students struggle with learning formulas in math mainly because they have trouble being engaged. Students should want to be more engaged in learning math. Ways to make student more engaged is by giving students hands on projects so that they have a goal in what they want to do. If students aren't engaged in class, then they will not learn the idea that is being presented to them. If we can get students to be engaged in class. They will not be distracted in class

Element A: Presentation and Justification of the Problem

	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
A1. Problem identification is _____.	very clear and objective	clear and objective	somewhat clear and objective	somewhat clear or objective	unclear	missing (cannot be inferred from information included)
A2. Problem definition is _____.	in considerable depth	in more than adequate depth	in adequate depth	in superficial depth	unclear	missing (cannot be inferred from information included)
A3. Problem elaboration is _____.	well elaborated	elaborated	imprecisely or generally elaborated	minimally elaborated	unelaborated and/or is clearly subjective	not elaborated (cannot be inferred from information included)
A4. Concern of primary stakeholder groups. Justification of the problem highlights the concerns of _____.	almost all primary stakeholder groups	many primary stakeholder groups	several primary stakeholder groups	at least a few primary stakeholders/stakeholder groups	only one or two stakeholders/stakeholder groups	no primary stakeholders/stakeholder groups
A5. Problem justification is based on _____.	numerous, timely, credible, and various sources	timely, credible, and various sources	credible and timely sources	credible sources	partially-credible sources	essentially only the opinion of the researcher
A6. Problem justification offers _____ -objective detail.	consistently	mostly	generally	somewhat	unclear	no